



# Shouvik Chaudhuri, Ph.D.

SMIEEE (US) | MIET (UK) | MIE (India) | Pursuing CEng status (IET)

## Researcher in Dynamics and Control

Nonlinear and Adaptive Control | Safety-Critical and Learning-Based Control | Maritime and Electrohydraulic Systems

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## Research Profile

I am a control engineer with more than twelve years of experience in the nonlinear and adaptive control of uncertain dynamical systems, and in the mathematical modelling that supports it. My work spans learning-based control, in which the learned component sits inside the loop under an adaptation law designed to preserve stability; safety-critical formulations built on control barrier functions and quadratic programming; and maritime motion control. The common thread is closed-loop behaviour that can be certified rather than only tuned, on plants that are uncertain and only partly modelled. I take that work from mathematical formulation through simulation and MIL/SIL/HIL pipelines to real-time experimental validation on purpose-built rigs.

That control track is the centre of my work. Around it sit three further strands: electrohydraulic actuation and parallel manipulators, which formed my doctoral research and remain an active collaboration; energy and thermal systems, through refrigeration and maritime energy work at SDU; and the engineering-biology interface, where I have contributed image processing and closed-loop control to peer-reviewed studies in molecular and cell biology and in biomedical device control. I have led a three-partner consortium as Project Manager, held Principal Investigator status on two competitive grants, and supervised twelve master's and bachelor's projects.

## Technical Skills

**Control theory and algorithms:** adaptive and nonlinear control; safety-critical control using control barrier functions and QP safety filters; model predictive control; sliding mode control; neural-network and fuzzy control; learning-based and reinforcement-learning control; robust and optimal control.

**Modelling and simulation:** nonlinear dynamic modelling, model-based design, digital twins, parameter identification, Kalman filtering, state estimation and observer design.

**Application domains:** marine vessel motion control, roll stabilisation and seasickness reduction; electrohydraulic actuation and parallel manipulators; refrigeration and heat-pump systems; vehicle dynamics; robotics.

**Real-time implementation:** MIL, SIL and HIL pipelines; Speedgoat, NI cRIO and myRIO, DAQ targets; embedded platforms over CAN, SPI, I<sup>2</sup>C, UART and RS-485; IMUs, encoders, LVDTs, pressure and flow sensors, load cells; specification and commissioning of experimental test rigs.

**Actuation and drives:** high-performance servo, proportional and servo-proportional valves (Rexroth, Eaton, Moog); motor control and VFD; hydraulic test benches; Stewart platform.

**Tools:** MATLAB/Simulink including Simscape, LabVIEW and the NI toolchain, Automation Studio, L<sup>A</sup>T<sub>E</sub>X.

## Professional Experience

### Postdoctoral Researcher — Maritime Control Systems

03/2022 – 02/2026

*Centre for Industrial Mechanics (CIM), University of Southern Denmark (SDU)*

*Sønderborg, Denmark*

### SAFEMARVEL — Safety-critical control framework for roll stabilisation of marine vessels

*Funder: Den Danske Maritime Fond. Role: Project Manager and Research Lead. Consortium: SDU Sønderborg (R&D), Dacoma ApS (technology owner), SDU Physics Odense (academic partner).*

- Formulated, proved and validated a safety-critical closed-loop framework using control-barrier-function constraints to guarantee that the plant state remains within a defined safe set under bounded disturbance, with the performance and constraint trade-off resolved by a quadratic program solved at every sample.
- Designed the nominal high-performance controllers and the QP safety filter mediating between them, including a learning-based controller with particular focus on reinforcement learning across varying sea states.
- Verified closed-loop behaviour through MIL/SIL/HIL pipelines and a purpose-built wave-generating test rig that I specified and commissioned.

- Owned the project management work package: planning and chairing consortium meetings, technical resource forecasting, estimating time-to-complete, funder reporting, and aligning academic and industrial timelines.
- Owned the dissemination work package: peer-reviewed journal articles, international conferences and stakeholder outreach to the wider maritime sector.

**AMCOSTAR** — *Dynamic Airkeel stabiliser for increasing the capacity of smaller boats through active control*

*Funder: Eurostars/Eureka Network and Innovation Fund Denmark. Role: Researcher and Project Participant. Consortium: Dacoma ApS, TUCO Marine Group, Automasjon & Data, SDU Sønderborg.*

- Led development and validation of nonlinear feedback control strategies regulating vessel motion against wave-induced disturbances across varying sea states.
- Integrated control algorithms, sensors and actuators with physical hardware through simulation, digital twin validation and experimental testing.
- Demonstrated the stabilising and load-carrying capability of an integrated Airkeel and adaptive control system on a small boat.
- Collaborated across a four-organisation, two-country consortium spanning control design, vessel manufacture, hardware development and data-driven system integration, and directed technical reporting and stakeholder engagement.

**Visiting Researcher (Secondment) — PUREWATER Project**

06/2023

*Control Systems Group, Institute of Automation and Systems Engineering, TU Ilmenau Ilmenau, Germany*

- Seconded from SDU Mechatronics under the EU Marie Skłodowska-Curie Actions (MSCA) project PUREWATER.
- Conducted collaborative research with the host group (Prof. Johann Reger) and industry partner KOMPASS GmbH on automation and systems engineering for water-related applications.

**Research Fellow — Robotics and Electrohydraulic Control Systems**

08/2013 – 02/2022

*Hydraulics Laboratory, Dept. of Mechanical Engineering, Jadavpur University*

*Kolkata, India*

**CARS** — *Design and development of a quadruped torso with an external electrohydraulic power system for an autonomous robot*

*Sponsor: Centre for Artificial Intelligence & Robotics (CAIR), DRDO, Bengaluru. Principal Investigator: Prof. Saikat Mookherjee. Role: Researcher and Technical Contributor. 12/2015 – 02/2022.*

- Designed and implemented real-time, multi-actuator control architectures in LabVIEW, including control logic and feedback loops, within an interdisciplinary team.
- Conducted mathematical modelling, simulation and genetic-algorithm-based optimisation studies in MATLAB/Simulink.
- Led system integration, testing and experimental validation; managed laboratory infrastructure and equipment procurement.

**DARO** — *High-frequency real-time tracking control for a linear servo actuation system*

*Sponsor: Aeronautics Research & Development Board (AR&DB), DRDO. Principal Investigator: Prof. Rana Saha. Role: Researcher and Technical Contributor. 08/2013 – 12/2015.*

- Developed adaptive and model-free strategies for high-frequency motion control of linear actuators, using sliding-mode, fuzzy and recurrent-neural-network methods under hard nonlinearities and parametric drift.
- Built and validated a real-time vision-based motion sensing and tracking system on a 700 kg hydraulic Stewart platform, published in the *IEEE Sensors Journal*.
- Modelled and simulated the system in MATLAB/Simulink, with real-time validation in LabVIEW.

**Masters GATE Fellow**

07/2012 – 06/2013

*Instrumentation & Control Division, Bhabha Atomic Research Centre (BARC)*

*Mumbai, India*

- Built a first-principles MATLAB/Simulink model of a pressurised water reactor with primary and secondary loops, together with its reactor regulating system.
- Assessed model stability against the Lyapunov criteria; the work was later published as a book.

## Research Funding and Grants

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### Funded projects

1. *Safety-critical control framework for roll stabilisation of marine vessels (SAFEMARVEL)*. Den Danske Maritime Fond, grant 2023-116 / 3410157, 03/2024 – 02/2026. **Role:** Research Lead and Project Manager; ideation and grant-preparation lead. **Amount:** 1,489,000 DKK (approx. €200,000). **Status:** Completed.
2. *Empowering engineering education at SDU with an advanced servo motor control system*. Fabrikant Mads Clausen Fond, grant 3410254, 01/2025 – 02/2026. **Role:** Principal Investigator. **Amount:** 100,000 DKK (approx. €13,500). **Status:** Completed. Specified, procured, commissioned and integrated a Speedgoat Baseline real-time target with FPGA-based configurable I/O and igus Drylin precision linear stages into the Cyber-Physical Laboratory at SDU Sønderborg, establishing a permanent testbed for real-time implementation of PID, feedforward and nonlinear motion-control strategies.
3. *Refrigeration, air-conditioning and heat-pump (RACHP) training system for next-generation Danfoss engineers at SDU*. Fabrikant Mads Clausen Fond, grant 3410089, 01/2024 – 12/2025. **Role:** Principal Investigator. **Amount:** 100,000 DKK (approx. €13,500). **Status:** Completed. Led specification, acquisition, commissioning and curricular integration of an inverter-compressor heat-pump trainer with electronic expansion valves, variable-speed fans and temperature, pressure and power instrumentation.

### Travel and mobility support

1. *Otto Mønstedts Fond*, 2026. Travel support for conference participation in INCOM 2026, India. **Role:** Main applicant. **Amount:** 7,500 DKK (approx. €1,000).

### Proposals submitted, under review or not funded

1. *MERLIN: Marine Environmental Remediation Learning Integrated Navigator*. Villum Foundation, Villum Experiment call, 2026. **Role:** Lead applicant. **Status:** Progressed to Phase 2.
2. *Smart energy management and power flow optimisation for maritime energy hubs*. EUDP-II, September 2026 call. **Role:** Project participant. **Status:** Under preparation.
3. *Smart power flow optimisation for megawatt supercharging in maritime onshore power supply*. Energy Cluster Denmark, Business Lighthouse South and North, Phase III, 2025. **Role:** Proposal contributor and co-developer. **Status:** Waitlisted.
4. *Motion-stabilised safe room for reducing motion sickness on marine vessels*. Den Danske Maritime Fond, 2025. **Role:** Proposal contributor and co-developer. **Status:** Not funded.

## Educational Qualifications

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### Ph.D. (Engineering)

07/2015 – 11/2021

Department of Electrical Engineering, Jadavpur University, Kolkata, India CGPA 10.00/10.00 | EQF Level 8

- **Thesis:** *Developing Electrohydraulic System Solutions using Adaptive-Neuro-Sliding Mode Control and Vision Sensing*.
- **Supervisors:** Prof. Amitava Chatterjee (Electrical Engineering) and Prof. Saikat Mookherjee (Mechanical Engineering).
- **Fields:** adaptive control, sliding mode control, neural networks, image-based visual servoing, electrohydraulic actuation systems, parallel manipulators.

### M.E. in Nuclear Engineering

08/2011 – 07/2013

Jadavpur University, Kolkata, India

CGPA 9.02/10.00 (86.28%) | EQF Level 7

- **Thesis:** *Compact Pressurized Water Reactor Modelling*, carried out at BARC Mumbai.
- **Major subjects:** Reactor Physics and Engineering I & II, Reactor Control Engineering, Concepts in Nuclear Science, Active Circuits and Systems, Reactor Thermal Hydraulics, Microscale Heat Transfer, Two-Phase Flow.

### B.Tech in Electrical Engineering

08/2007 – 06/2011

Seacom Engineering College, West Bengal University of Technology, India

DGPA 8.73/10.00 | EQF Level 6

- **Major subjects:** Control Systems I & II, Electrical Machines I & II, Electromagnetic Field Theory, Microprocessors and Microcontrollers, Digital Signal Processing.

### Higher Secondary (Class XII), CBSE

April 2006

- **Subjects:** Physics, Chemistry, Mathematics, Biology, English.

**Secondary (Class X), CBSE**

April 2004

Kendriya Vidyalaya Ballygunge, Kolkata, India

Aggregate 90.60%

- **Subjects:** Science, Mathematics, Social Science, English, Hindi.

**Dissertations**

1. **Doctoral dissertation**, 2015 – 2021, Jadavpur University. *Developing Electrohydraulic System Solutions using Adaptive-Neuro-Sliding Mode Control and Vision Sensing*. Real-time control solutions for industry-grade electrohydraulic actuation systems using hybrid adaptive strategies built around sliding mode control and recurrent neural networks, together with an image-based visual servoing motion-sensing application on a multi-actuator parallel manipulator (Stewart platform) with a monocular eye-in-hand camera.
2. **Postgraduate dissertation**, 07/2012 – 06/2013, Bhabha Atomic Research Centre (BARC), Mumbai. *Compact Reactor Modelling*. Advisor: Dr. Siddhartha Mukhopadhyay (Scientist H+, I&C Division). Fields: pressurised water nuclear reactors, reactor modelling, thermal hydraulics, xenon poisoning, reactor regulating systems, Lyapunov stability.
3. **Undergraduate dissertation**, 12/2010 – 04/2011, Seacom Engineering College, WBUT. *Study and Fabrication of a Battery Low Indicator*. Advisor: Mr. Sarbojit Mukherjee. Fields: circuit simulation and fabrication, DC and transient analysis.

**Publications**

*Summary:* 14 peer-reviewed journal papers | 11 conference proceedings | 1 book | 2 book chapters.

*Bibliometrics:* h-index 7. Web of Science (September 2026): 122 citations across 25 indexed publications, 17 in the Core Collection. Google Scholar: 184 citations.

**Journal papers (reverse chronological)**

1. V. Joseph, B.K. Sarkar, A. Chatterjee, **S. Chaudhuri**, “Fast nonsingular terminal sliding mode controller design for Francis turbine with electrohydraulic IGV: an experimental investigation,” *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, Jan. 2026. DOI: [10.1177/09596518251399972](https://doi.org/10.1177/09596518251399972).
2. Y. Sheikh Kilo, R. Izadi-Zamanabadi, H. Ramezani, P. Gullo, **S. Chaudhuri**, “Cost-effective optimization for charging/discharging cycles of thermal energy ice storages in transcritical R744 supermarket refrigeration systems,” *Energy Conversion and Management*, vol. 334, 119790, Jun. 2025. ISSN 0196-8904. DOI: [10.1016/j.enconman.2025.119790](https://doi.org/10.1016/j.enconman.2025.119790).
3. S. Paul, S. Das, M. Banerjee, **S. Chaudhuri**, B. Das, “The ATP-dependent DEAD-box RNA helicase Dbp2 regulates the glucose/nitrogen stress response in baker’s yeast by modulating reversible nuclear retention and decay of *SKS1* mRNA,” *Genetics*, vol. 229, no. 3, iyae221, Mar. 2025. DOI: [10.1093/genetics/iyae221](https://doi.org/10.1093/genetics/iyae221).
4. H. Ramezani, **S. Chaudhuri**, J. Jouffroy, A. Baurichter, S.M. Hansen, “On roll stabilisation using a canting keel,” *IEEE Transactions on Control Systems Technology*, vol. 33, no. 2, pp. 539–553, Mar. 2025 (early access Nov. 2024). ISSN 1558-0865. DOI: [10.1109/TCST.2024.3494237](https://doi.org/10.1109/TCST.2024.3494237).
5. N. Taghva, **S. Chaudhuri**, “How language-specific and cross-linguistic factors affect speech rhythm,” *Estudios de Fonética Experimental*, vol. 33, pp. 169–190, Sep. 2024. DOI: [10.1344/EFE-2024-33-169-190](https://doi.org/10.1344/EFE-2024-33-169-190).
6. A. Sarkar, S. Dasmahapatra, **S. Chaudhuri**, R. Saha, S. Mookherjee, D. Sanyal, “A novel order-separated generalized feedforward design for motion control in energy-efficient electrohydraulic systems with proportional and integral feedback,” *ISA Transactions*, vol. 146, pp. 336–351, Jan. 2024. ISSN 0019-0578. DOI: [10.1016/j.isatra.2024.01.012](https://doi.org/10.1016/j.isatra.2024.01.012).
7. I. De, S. Sarkar, **S. Chaudhuri**, N. Mondal, N. Kumar, “Effect of dynamic swivelling torque and eccentricity on the design of compensator cylinders for a variable displacement axial piston pump,” *Jordan Journal of Mechanical and Industrial Engineering*, vol. 17, no. 2, pp. 255–268, Jun. 2023. ISSN 1995-6665. DOI: [10.59038/jjmie/170209](https://doi.org/10.59038/jjmie/170209).
8. A. Sarkar, K. Maji, **S. Chaudhuri**, R. Saha, S. Mookherjee, D. Sanyal, “Actuation of an electrohydraulic manipulator with a novel feedforward compensation scheme and PID feedback in servo-proportional valves,” *Control Engineering Practice*, vol. 135, 105490, Mar. 2023. ISSN 0967-0661. DOI: [10.1016/j.conengprac.2023.105490](https://doi.org/10.1016/j.conengprac.2023.105490).

9. S. Sarkar, R. Mandal, N. Mondal, **S. Chaudhuri**, T. Mandal, G. Majumdar, "Modelling and prediction of micro-hardness of electroless Ni-P coatings using response surface methodology and fuzzy logic," *Jordan Journal of Mechanical and Industrial Engineering*, vol. 16, no. 5, pp. 729–742, Dec. 2022. ISSN 1995-6665.
10. P. Datta, A. Chakravarty, R. Saha, **S. Chaudhuri**, K. Ghosh, A. Mukhopadhyay, S. Sen, A. Dutta, P. Goyal, "Experimental investigation on the effect of initial pressure conditions during steam-water direct contact condensation in a horizontal pipe geometry," *International Communications in Heat and Mass Transfer*, vol. 121, 105082, Feb. 2021. ISSN 0735-1933. DOI: [10.1016/j.icheatmasstransfer.2020.105082](https://doi.org/10.1016/j.icheatmasstransfer.2020.105082).
11. **S. Chaudhuri**, R. Saha, A. Chatterjee, S. Mookherjee, D. Sanyal, "Adaptive neural-bias-sliding mode control of rugged electrohydraulic system motion by recurrent Hermite neural network," *Control Engineering Practice*, vol. 103, 104588, Oct. 2020. ISSN 0967-0661. DOI: [10.1016/j.conengprac.2020.104588](https://doi.org/10.1016/j.conengprac.2020.104588).
12. **S. Chaudhuri**, R. Saha, A. Chatterjee, S. Mookherjee, D. Sanyal, "Development of a motion sensing system based on visual servoing of an eye-in-hand electrohydraulic parallel manipulator," *IEEE Sensors Journal*, vol. 20, no. 14, pp. 8108–8116, Jul. 2020. ISSN 1558-1748. DOI: [10.1109/JSEN.2020.2979490](https://doi.org/10.1109/JSEN.2020.2979490).
13. S. Sarkar, A. Mukherjee, S. Das, B. Ghosh, **S. Chaudhuri**, D. Bhattacharya, A. Sarbajna, R. Gachhui, "Nitrogen deprivation elicits dimorphism, capsule biosynthesis and autophagy in *Papiliotrema laurentii* strain RY1," *Micron*, vol. 124, 102708, Sep. 2019. ISSN 0968-4328. DOI: [10.1016/j.micron.2019.102708](https://doi.org/10.1016/j.micron.2019.102708).
14. S. Das, S. Biswas, **S. Chaudhuri**, A. Bhattacharyya, B. Das, "A nuclear zip code in *SKS1* mRNA promotes its slow export, nuclear retention, and degradation by the nuclear exosome/DRN in *Saccharomyces cerevisiae*," *Journal of Molecular Biology*, vol. 431, no. 19, pp. 3626–3646, Sep. 2019. ISSN 0022-2836. DOI: [10.1016/j.jmb.2019.07.005](https://doi.org/10.1016/j.jmb.2019.07.005).

#### Conference proceedings (reverse chronological)

1. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, "Ensuring safe roll dynamics in marine vessels based on a safety-enhanced control framework," *International Conference on Electrical, Computer and Energy Technologies (ICECET)*, 2026. Presented.
2. Z. Gniewosz, **S. Chaudhuri**, H. Ramezani, "Torque vectoring-based active yaw control for enhanced vehicle handling and stability using a digital twin framework," *INCOM 2026*, Kolkata, India, Jan. 2026. Presented.
3. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, "Remarks on augmenting stability and safety in nominal controllers: a practical perspective," *European Control Conference (ECC)*, Thessaloniki, Greece, 2025. DOI: [10.23919/ECC65951.2025.11187095](https://doi.org/10.23919/ECC65951.2025.11187095).
4. H. Ramezani, **S. Chaudhuri**, J. Jouffroy, A. Baurichter, S.M. Hansen, "On roll stabilisation using a canting keel," *62nd IEEE Conference on Decision and Control (CDC)*, Singapore, Dec. 2023. DOI: [10.1109/CDC49753.2023.10384049](https://doi.org/10.1109/CDC49753.2023.10384049).
5. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, "Investigation of marine vessels equipped with an airkeel subject to wave-induced disturbances," *IEEE Conference on Control Technology and Applications (CCTA)*, Bridgetown, Barbados, Aug. 2023. DOI: [10.1109/CCTA54093.2023.10252537](https://doi.org/10.1109/CCTA54093.2023.10252537).
6. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, "Modelling and control of a canting-keel based ship roll stabilization system for crane operations," *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Prague, Czech Republic, Oct. 2022. DOI: [10.1109/SMC53654.2022.9945570](https://doi.org/10.1109/SMC53654.2022.9945570).
7. R. Bhattacharjee, S. Kundu, **S. Chaudhuri**, "Evaluation of workspace and coupled motions of an electrohydraulic parallel manipulator," *Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT)*, Erode, India, Sep. 2021. DOI: [10.1109/ICECCT52121.2021.9616672](https://doi.org/10.1109/ICECCT52121.2021.9616672).
8. S. Kundu, R. Bhattacharjee, **S. Chaudhuri**, "Evaluation of fuzzy-logic based position control strategies for an electrohydraulic actuation system," *International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, Bhilai, India, Feb. 2021. DOI: [10.1109/ICAECT49130.2021.9392479](https://doi.org/10.1109/ICAECT49130.2021.9392479).
9. **S. Chaudhuri**, R. Saha, S. Mookherjee, D. Sanyal, A. Chatterjee, "Visual sensing based adaptive sliding mode control of position tracking in electrohydraulic systems," *Second International Conference on Control, Instrumentation, Energy & Communication (CIEC)*, Kolkata, India, Jan. 2016. DOI: [10.1109/CIEC.2016.7513762](https://doi.org/10.1109/CIEC.2016.7513762).
10. **S. Chaudhuri**, S. Mookherjee, D. Sanyal, "Adaptive force tracking in electrohydraulic system with first-order sliding mode control," *IEEE First International Conference on Control, Measurement and Instrumentation (CMI)*, Kolkata, India, Jan. 2016. DOI: [10.1109/CMI.2016.7413759](https://doi.org/10.1109/CMI.2016.7413759).
11. S. Dasmahapatra, **S. Chaudhuri**, P. Mandal, S. Mookherjee, R. Saha, "Fuzzy-PI control of motion tracking by an electrohydraulic system with multiple nonlinearities," *IET Michael Faraday International Summit*, Kolkata, India, Sep. 2015. DOI: [10.1049/cp.2015.1612](https://doi.org/10.1049/cp.2015.1612).

#### Book

1. **S. Chaudhuri**, *Pressurised Water Nuclear Reactor: Dynamics, Modelling and Simulation*, LAP Lambert Academic Publishing (OmniScriptum Publishing Group), 164 pp., 2019. ISBN 978-620-0-09362-2.

## Book chapters

1. R. Bhattacharjee, **S. Chaudhuri**, A. Ganguly, “Robust control of pulsatile ventricular assist devices for patients with advanced heart failure,” in *Proceedings of the International Conference on Frontiers in Computing and Systems (COMSYS 2022)*, Lecture Notes in Networks and Systems, Springer, Singapore, 2023, pp. 331–341. ISBN 978-981-99-2679-4. DOI: [10.1007/978-981-99-2680-0\\_29](https://doi.org/10.1007/978-981-99-2680-0_29).
2. **S. Chaudhuri**, S. Dasmahapatra, A. Chatterjee, R. Saha, S. Mookherjee, D. Sanyal, “Adaptive fuzzy-sliding mode control with fixed bias compensator for an electrohydraulic actuation system with hard nonlinearities,” in *Fluid Mechanics and Fluid Power: Contemporary Research*, Lecture Notes in Mechanical Engineering, Springer, New Delhi, 2017, pp. 1223–1232. ISBN 978-81-322-2743-4. DOI: [10.1007/978-81-322-2743-4\\_116](https://doi.org/10.1007/978-81-322-2743-4_116).

## Teaching Experience

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### Main lecturer or co-lecturer, University of Southern Denmark

- **Experimental Control Systems (XCOS)** | MSc Mechatronics, 2nd semester | 5.0 ECTS | Spring 2023, 2024, 2025 | *Role: Lecturer*.  
Practical and experimental application of control systems techniques. Students implement real-time controllers on physical rigs including inverted pendulums, helicopter models and multi-tank systems, integrating MATLAB/Simulink modelling with sensor feedback and DAQ-based implementation.
- **Expert in Teams (EXT)** | BSc, all disciplines, 5th semester | 10.0 ECTS | Fall 2023, 2024, 2025 | *Role: Teacher and Supervisor*.  
Interdisciplinary teams drawn from Electronics, Mechatronics, Mechanical, Software Engineering and Engineering Innovation work on stakeholder-defined industrial projects from problem formulation through prototype to business case.
- **Mechanical Semester Project, Hydraulic Power Systems (SPRO4ME)** | BSc Mechanical, 4th semester | 10.0 ECTS | Spring 2025 | *Role: Co-lecturer*.  
Project-based course in fluid mechanics and hydraulic power systems. Students design, simulate and dimension pumps, cylinders and control valves, applying foundational control technology and international standards in a multidisciplinary team context. Includes an industry visit to the Danfoss main campus.
- **Control Engineering 1 (CoE1)** | BSc, all disciplines, 4th semester | 5.0 ECTS | Spring 2024 | *Role: Co-lecturer*.  
Introductory control systems: system modelling, signal flow graphs, Routh stability criteria, root locus, frequency response techniques and PID control.

### Teaching assistant

- **Control of Autonomous Systems (CAS)** | MSc Mechatronics | 5.0 ECTS | Autumn 2022.
- **Construct Mechatronics** | BSc | 10.0 ECTS | Spring 2022.

### Examination

- Internal co-examiner at SDU for master-level courses including Adaptive and Nonlinear Control, Fault-Tolerant Control, and Statistical Signal Processing.

## Student Supervision

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### Master’s (postgraduate) dissertations supervised: 6

1. *Flow-Controlled Underfloor Heating Using PCM-Assisted Flow Wall Systems* — Bragi Sigurjónsson, 2026.
2. *Amfitrack EMF Data Fusion* — Frej Karlinsky Scherfig, 2025.
3. *Development of Trajectory Generation Algorithms for CNC Foam Cutting Machines* — Arnas Serva, 2025.
4. *Autonomous Aerial Mapping and SLAM: A Drone-Based Approach for 3D Environmental Reconstruction* — David Milošević, 2025.
5. *Design and Development of a Diesel Dosing Unit for Active Regeneration of DPF Filters in Exhaust After-Treatment Systems* — Arfeen Ahmed Ali, 2025.
6. *Verification and Validation of Custom Code Generation from Model-Based Design using Simulink* — Jacob Thomas Puthukeril, 2023.

### Bachelor’s (undergraduate) projects supervised: 6

1. *Development of a Simulation-Based Sales Tool: Quantifying Efficiency Gains of PVG Nova vs. PVG 32* — Jana Aly and Jakub Edward Brodzinski, 2026.
2. *Modelling and Control of a Time-Pressure Membrane Valve System for Pharmaceutical Filling* — Einar Benjamin Jensen, 2026.

3. *Modelling and Control of Propeller-Drive Systems with Nonlinear Dynamics for Marine Applications* — Laura Zanón Barney, 2026.
4. *Orientation Estimation and Control for Thruster-Assisted Wind Turbine Installation* — Henrik Maarten Bongers, 2026.
5. *Active Yaw Control for Vehicle Handling and Stability with Torque Vectoring* — Zofia Roza Gniewosz, 2025.
6. *Design and Development of a Semi-Active Magnetorheological Suspension for Automotive and Motorsport Applications* — Lucas Fernández Schelstraete, 2024.

Several projects ran with industry partners and produced peer-reviewed publications with the student as lead author, including INCOM 2026. I also provided day-to-day scientific guidance to junior researchers on the SAFE-MARVEL and AMCOSTAR teams.

## Membership in Professional Bodies

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- **Senior Member (SMIEEE)**, Institute of Electrical and Electronics Engineers (IEEE), US. Elevated 2026; member since 2016. Member ID 90902393. Kolkata Section.
- **Member (MIET)**, Institution of Engineering and Technology (IET), UK. Member ID 1101020475. Pursuing CEng status.
- **Member (MIE) and Chartered Engineer (India)**, Institution of Engineers (IE(I)), India. Member ID M-1848040.
- **Associate Member**, Indian National Academy of Engineering (INAE), India. Application under review, 2026.

## Editorial and Peer-Review Activities

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**55 verified reviews of 42 manuscripts** (September 2015 – September 2026), recorded under Web of Science ResearcherID [S-9653-2019](#).

- **IEEE/ASME Transactions on Mechatronics** — 14 reviews.
- **IEEE Transactions on Instrumentation and Measurement** — 10 reviews.
- **IEEE Transactions on Vehicular Technology** — 9 reviews.
- **IEEE International Instrumentation and Measurement Technology Conference (I2MTC)** — 6 reviews.
- **IEEE Sensors Journal** — 3 reviews.
- **Control Engineering Practice** (Elsevier) — 3 reviews.
- **Proceedings of the Institution of Mechanical Engineers, Part C** — 2 reviews.
- **Computers and Electronics in Agriculture** (Elsevier) — 2 reviews.
- **IEEE Journal of Emerging and Selected Topics in Industrial Electronics (JESTIE)** — 2 reviews.
- **Sādhana** (Springer) — 1 review.
- **Journal of the Brazilian Society of Mechanical Sciences and Engineering** (Springer) — 1 review.
- **IEEE Open Journal of Instrumentation and Measurement** — 1 review.
- **IEEE Industry Applications Society (IAS) Publications** — 1 review.

## Honours and Awards

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- Elevated to **Senior Member** grade of the IEEE, 2026.
- **International mobility and travel award**, Otto Mønstedts Fond, 2026.
- **Winner, IEEE COVID-19 App Development Contest** (IEEE Bangalore Section), 2020, for co-developing *CovCov*, a COVID-19 mobile application, as part of a Jadavpur University team.

## Key Achievements

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- Delivered safety-critical closed-loop control for active marine vessel motion stabilisation, validated across MIL, SIL and HIL pipelines and on experimental rigs.
- Specified and commissioned a marine vessel test rig with integrated wave generation for real-time controller validation under disturbance.
- Engineered a real-time, multi-actuator control architecture for an electrohydraulic quadruped robotic platform as part of an interdisciplinary team.

- Developed a real-time vision-based motion sensing and tracking system on a 700 kg hydraulic Stewart platform.
- Established two permanent teaching and research testbeds at SDU as Principal Investigator on competitive grants.

## Language Skills

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English (Professional, C1; certified by the British Council) | Hindi (Native, C2) | Bengali (Native, C2) | Danish (Basic, A1) | German (Beginner).

## References

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